

Fabrication and Electrical Characterization of HEMTs

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Course: Microsystems Fabrication Technology

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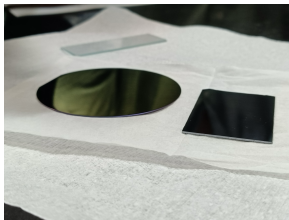
1. Introduction to HEMT & 2DEG Channel Physics

Gallium Nitride (GaN) and Aluminum Gallium Nitride ($\text{Al}_x\text{Ga}_{1-x}\text{N}$) possess exceptional physical properties for high-power and RF electronics, including wide direct bandgaps ($E_{g,\text{GaN}} \approx 3.4 \text{ eV}$, $E_{g,\text{AlN}} \approx 6.2 \text{ eV}$), high critical breakdown fields ($> 3.3 \text{ MV/cm}$), and high electron saturation velocity ($\sim 2.5 \times 10^7 \text{ cm/s}$). Unlike conventional silicon MOSFETs that rely on intentional impurity doping (which degrades mobility via ionized impurity scattering), the HEMT utilizes spontaneous (P_{SP}) and strain-induced piezoelectric (P_{PE}) polarization along the $[0001]$ c-axis. This polarization discontinuity produces an intense positive interface sheet charge ($+\sigma_{\text{pol}} \approx 10^{13} \text{ e/cm}^2$), inducing sharp downward band bending below the Fermi level (E_F). Electrons accumulate in the resulting triangular quantum well at the hetero-interface, forming a Two-Dimensional Electron Gas (2DEG) characterized by high sheet density ($n_s \sim 10^{13} \text{ cm}^{-2}$) and mobility ($\mu_n > 1500 \text{ cm}^2/\text{V}\cdot\text{s}$) without any intentional doping.

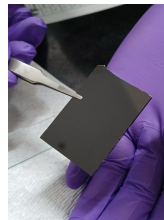
2. Substrate Inspection & Wafer Cleaving: HEMT vs. Silicon Distinctions

2.1 Visual Differences Between HEMT and Bulk Silicon Wafers

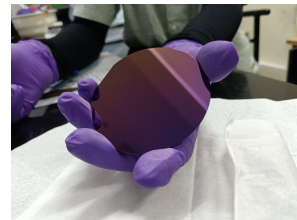
During the preliminary laboratory demonstration, students were shown both an HEMT wafer grown on a Silicon (111) substrate and an unpatterned bare monocrystalline Silicon wafer to examine their macroscopic structural and optical differences. The two wafers exhibited distinct, highly discernible visual characteristics:



(a)



(b)



(c)

Figure 1: (a) HEMT and Si wafer looking similar when viewed from the side

(b) HEMT wafer giving metallic greyish color when viewed from the top in a lighted environment

(c) Si wafer giving purple color when viewed from the top in a lighted environment

2.2 Wafer Cleaving and Slicing (Conducted on Silicon Wafer)

Crucially, to protect the fragile, high-cost epitaxial HEMT heterostructure wafer from accidental surface scratching or catastrophic fracture, wafer dicing and slicing practice was performed exclusively on the bulk Silicon wafer. The Si wafer was placed on a clean glass support carrier. A diamond-tipped precision scribe was used to slice the wafer. Applying gentle, controlled downward cantilever pressure against the glass slide edge propagated an instantaneous, atomically flat cleavage crack across the substrate.

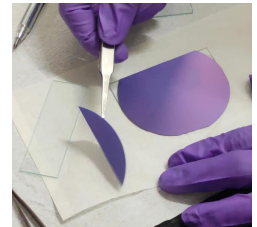


Figure 2: Si wafer slicing

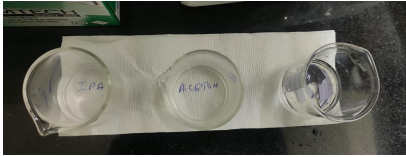
3. Substrate Degreasing & Surface Cleaning (RCA-Style Ultrasonic Protocol)

Surface cleanliness is paramount in microfabrication, as microscopic organic residues, airborne hydrocarbons, and ionic salts directly cause pinhole formation, interfacial voids, and severe film delamination during high-vacuum metallization. Furthermore, uncleaned organic films carbonize during subsequent high-temperature annealing, creating high-resistance parasitic conduction pathways. To achieve a pristine semiconductor surface, the diced substrates were subjected to a rigorous 4-stage sequential solvent degreasing protocol using a Rivotek Ultrasonic

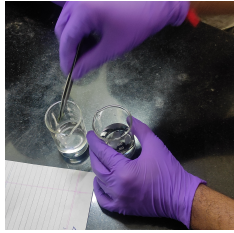
Cleaner bath. Ultrasonic agitation induces acoustic cavitation: high-frequency sound waves (> 30 kHz) generate millions of microscopic vapor cavities that violently implode, producing high-velocity micro-jets (pressures > 1000 atm, localized temperatures > 4000 K) that dislodge tenaciously bound organic contaminants without scratching the semiconductor surface.

Table 1: Sequential Substrate Degreasing and Surface Cleaning Protocol

Stage	Chemical Solvent	Bath Temperature	Duration	Physical & Chemical Rationale
1	Isopropanol (IPA)	Ambient (25°C)	10 min	Particulate removal, gross polar contamination stripping
2	Acetone (Heated)	$70^{\circ}\text{C} - 80^{\circ}\text{C}$	10 min	Aggressive dissolution of heavy oils, waxes, and organic grease
3	Methanol	Ambient (25°C)	10 min	Stripping of residual acetone films and ultra-fine polar traces
4	DI Water + N_2	Ambient	Cascade + Dry	Solvent displacement, ionic dilution rinse, inert drying



(a)



(b)



(c)

Figure 2: (a) Chemical solvents used in the substrate degreasing and surface cleaning protocol
(b) Specimen being treated with DI after dipping in each of the chemical solvents
(c) Inert drying the specimen using N_2 gas

4. Thermal Evaporation Metallization Stack (Al / Cr / Au)

Physical Vapor Deposition (PVD) of the multi-layer contact metallization was executed using an Advanced Process Technology high-vacuum box-coater thermal evaporator. The system was pumped down using a mechanical rotary vane roughing pump followed by a high-vacuum turbomolecular/diffusion pump to reach a base chamber pressure below 2.0×10^{-6} Torr. Operating at ultra-high vacuum ensures that the mean free path of evaporated metal atoms (~ 50 m) vastly exceeds the source-to-substrate distance (~ 30 cm), guaranteeing line-of-sight molecular trajectory with zero background gas scattering or oxidation.

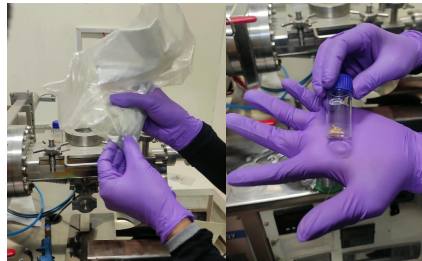
1. Aluminum (Al) — 180 nm (Primary Contact Layer): Directly contacts the semiconductor. Low work function (4.28 eV); reacts with GaN during annealing to liberate nitrogen and form donor vacancies (VN).

2. Chromium (Cr) — 30 nm (Diffusion Barrier & Adhesion Promoter): High melting point (1907°C); acts as a sacrificial barrier preventing Au from diffusing into GaN and spiking the 2DEG, while adhering Al to Au.

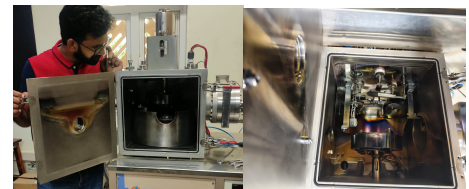
3. Gold (Au) — 200 nm (Low-Resistivity Capping & Probing Layer): Protects reactive Al/Cr from oxidation, provides excellent bulk conductivity, and forms a ductile pad for tungsten probe tip landing.



Figure 3: (a) Ultrasonic processor



(b) Sample handling



(c) Thermal evaporator chamber

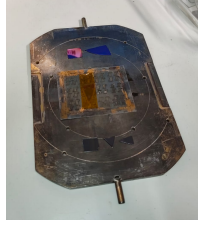
5. Photolithography & Resist Pattern Transfer

Photolithography was deployed to translate the microscale Source and Drain contact geometries from a high-contrast chrome-on-glass photomask into the deposited continuous 410 nm Al/Cr/Au metallization stack. The complete lithographic sequence followed precise thermal and rotational recipes:

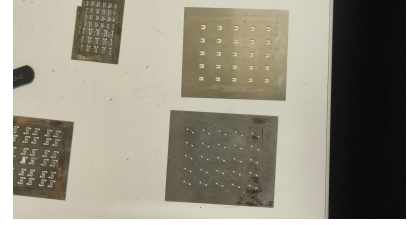
- **Spin Coating (Holmarc HO-TH-05C):** Substrate centered on vacuum chuck; positive diazonaphthoquinone (DNQ)-novolac resist dispensed. Accelerated at 500 rpm/s to a verified speed of 3915 rpm (~4000 rpm nominal) for 45 s, establishing uniform resist thickness of ~1.3 μm .
- **Soft Bake (Hot Plate, 115°C, 60 s):** Evaporates casting solvents (PGMEA), relieves internal film shear stress, and enhances resist-metal adhesion.
- **UV Mask Alignment & Exposure:** Substrate loaded into a contact mask aligner under a glass photomask bearing interdigitated HEMT geometries. Broadband UV exposure (365–436 nm) converted insoluble DNQ inhibitor in exposed regions into soluble indene carboxylic acid.
- **Development & Hard Bake (Hot Plate, 90°C, 90 s):** Immersed in positive developer for ~40 s, cleanly dissolving exposed resist, quenched in DI water, and baked at 90°C for 90 s to cross-link the resist matrix for acid etch resistance.



Figure 4: (a) PR spin coater



(b) Substrate Sample Holder & Shadow Mask Fixture Plate



(c) Metal hard shadow masks (stencil masks)

6. Selective Tri-Metal Wet Etching Chemistry (Au, Cr, Al)

Etching a trilayer metal stack (Au / Cr / Al) represents a formidable chemical engineering challenge due to the immense divergence in standard reduction potentials. A single universal etchant cannot be employed because electrochemical galvanic coupling would cause uncontrolled lateral undercut of the less noble Al layer. Therefore, an orthogonal, highly selective sequential wet chemical etching sequence was executed:

Table 3: Selective Tri-Metal Wet Etching Chemistry and Parameters

Etch Step	Target Metal	Chemical Reagents	Duration	Governing Chemical Mechanism
Step 1	Au (200 nm)	KI : I ₂ : H ₂ O (Iodine/Iodide)	10 – 20 s	$2\text{Au} + \text{I}_3^- + \text{I}^- \rightarrow 2[\text{AuI}_2]^-$ (Tri-iodide complexation; stops at Cr)
Step 2	Cr (30 nm)	(NH ₄) ₂ Ce(NO ₃) ₆ / HClO ₄	50 – 70 s	$\text{Ce}^{4+} + \text{Cr} \rightarrow \text{Ce}^{3+} + \text{Cr}^{6+}$ (Dichromate oxidation; stops at Al)
Step 3	Al (180 nm)	H ₃ PO ₄ : HNO ₃ : CH ₃ COOH : H ₂ O	4 × 30 s (120 s)	$\text{Al} + \text{HNO}_3 \rightarrow \text{Al}_2\text{O}_3$; $\text{Al}_2\text{O}_3 + 6\text{H}_3\text{PO}_4 \rightarrow 2\text{Al}(\text{H}_2\text{PO}_4)_3$ (Step-wise anti-undercut)

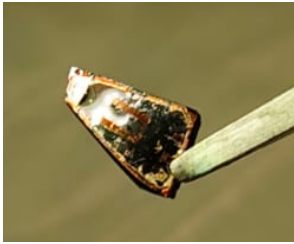
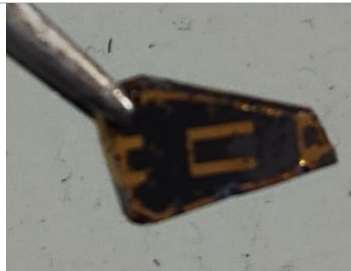
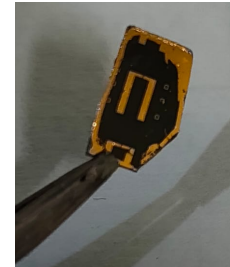


Figure 5: (a) Incomplete etching with resist scumming / residue



(b) Severe over-etching & pattern degradation



(c) Optimized selective wet etching

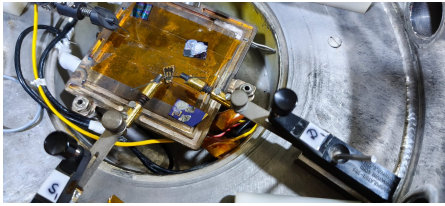
7. Electrical Characterization Setup (Keithley 6430 SMU & Probe Station)

Current-voltage (I–V) electrical characterization was conducted using an analytical manual probe station integrated with a Keithley 6430 Sub-Femtoamp Remote SourceMeter unit. The Keithley 6430 features a dedicated Remote PreAmp that resides within inches of the device under test, eliminating microphonic noise, dielectric absorption, and coaxial cable parasitic capacitance, thereby delivering ultra-sensitive current resolution (< 0.1 fA). Precision micro-manipulators carrying fine tungsten needle probes designated 'S' (Source) and 'D' (Drain) were landed onto the fabricated contact pads under optical stereo-microscopy. Data acquisition was automated using Keithley KickStart software, configured for bipolar and unipolar DC voltage sweeps with programmable current compliance limits.

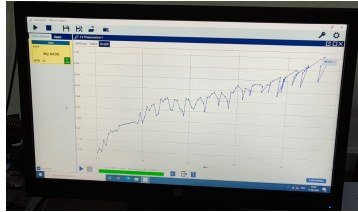
8. Pre-RTA Electrical Measurements (Open-Circuit / Schottky Barrier)

Initial electrical measurements performed immediately after wet etching and photoresist stripping (as-deposited state) revealed completely insulating, open-circuit behavior. As documented in the Keithley KickStart screen logs:

- **KickStart Voltage Sweep (0 V to 6 V):** The current response remained completely flat along the zero axis, fluctuating erratically within -50 pA to $+350$ pA. This is the background instrument noise floor.
- **Secondary Sweep (0 V to 5 V):** Current registered at low nanoampere levels (< 7 nA), confirming zero carrier transport across the contact pads.
- **Physical Origin of Non-Conduction:** As-deposited Aluminum on wide-bandgap AlGaIn forms a high rectifying Schottky barrier ($\Phi_B \approx 0.8\text{--}1.0$ eV). Furthermore, the wide-bandgap AlGaIn barrier layer (~ 25 nm) physically separates the metal from the buried 2DEG. Without thermal activation, thermionic emission over the barrier is exponentially suppressed, while the wide depletion width completely blocks quantum mechanical tunneling. Consequently, the device behaves as two back-to-back blocking Schottky diodes in series with an open circuit.



(a) Probe station contacting 'S' and 'D' pads



(b) Pre-RTA KickStart I–V curve (pA noise floor)



(c) Pre-RTA secondary sweep (low nA floor)

Figure 6: (a) Analytical probe station with micro-manipulator probes contacting Source ('S') and Drain ('D') pads. (b) Pre-RTA Keithley KickStart I–V characteristic exhibiting open-circuit behavior with current fluctuating in the picoampere noise floor (-50 pA to $+350$ pA). (c) Secondary Pre-RTA sweep confirming complete lack of carrier transport before contact alloying.

9. Rapid Thermal Annealing (RTA) & Contact Metallurgy Physics

To transform the high-resistance, rectifying as-deposited contacts into low-resistance ohmic contacts, the sample was processed in a Rapid Thermal Annealing (RTA) furnace at 850°C for 45 s under an ultra-pure N_2 ambient. The fundamental solid-state metallurgical reactions that govern ohmic contact formation to the buried 2DEG are as:

1. Nitrogen Extraction & Vacancy Generation (V_N^{++}): At 850°C , Aluminum undergoes an exothermic solid-state reaction with the GaN/AlGaIn crystal lattice: $\text{Al} + \text{GaN} \rightarrow \text{AlN} + \text{Ga} + \text{V}_\text{N}^{++}$. Because the standard Gibbs formation enthalpy of AlN ($\Delta H_f = -318$ kJ/mol) is dramatically more negative than that of GaN ($\Delta H_f = -110$ kJ/mol), Al aggressively extracts nitrogen from the sub-surface lattice. The liberated nitrogen leaves behind an extraordinary concentration of nitrogen vacancies (V_N) directly beneath the metal contact.

2. Formation of Degenerate n^{++} Tunneling Layer: In III-nitride semiconductors, nitrogen vacancies behave as shallow donor states. The dense accumulation of V_N defects creates an ultra-heavily doped degenerate n^{++} region where electron concentration exceeds 10^{19} cm^{-3} .

3. Depletion Barrier Thinning & Field Emission (FE): The depletion barrier thickness (W_D) at the metal-semiconductor interface is inversely proportional to the square root of donor doping: $W_D = \sqrt{[2 \epsilon_s (\Phi_B - V) / (q N_D)]}$. As N_D surges from $\sim 10^{16} \text{ cm}^{-3}$ to $> 10^{19} \text{ cm}^{-3}$, the effective barrier width collapses from tens of nanometers down to $<$

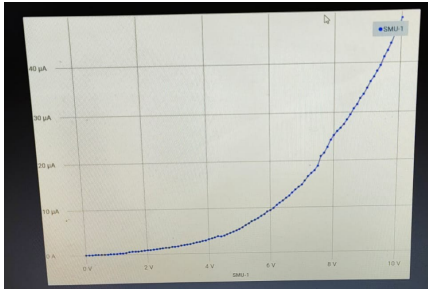
2 nm. This barrier thinning triggers Field Emission (quantum mechanical Fowler-Nordheim tunneling), enabling electrons to tunnel freely between the metal contact and the 2DEG channel with minimal contact resistance.

4. Diffusion Barrier Role of Chromium & Intermetallic Alloying: Chromium acts as an essential thermal diffusion barrier, preventing Gold from diffusing rapidly down into the GaN channel where it would cause destructive spiking and short-circuit the thin 2DEG. Concurrently, controlled intermixing produces Au-Al intermetallic phases (such as AuAl_2 'purple plague' and Au_2Al) that exhibit high electrical conductivity and thermal robustness.

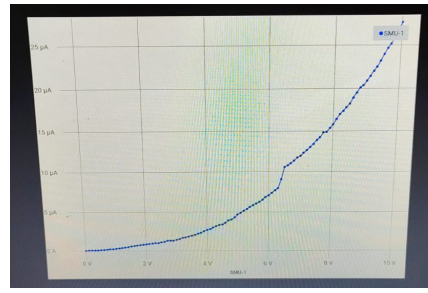
10. Post-RTA Electrical Characterization & Ohmic Activation

Following rapid thermal annealing at 850°C for 45 s, the devices were re-probed under identical bias configurations on the Keithley 6430 SMU. The measured I–V characteristics demonstrated an immediate, dramatic transition from open-circuit insulation to active carrier conduction across the 2DEG channel:

- **Device 1 Characterization (KickStart Log):** Current increases smoothly and monotonically with voltage, reaching a substantial $48\ \mu\text{A}$ at 10 V (compared to $< 0.35\ \text{nA}$ before annealing — an increase of over five orders of magnitude).
- **Device 2 Characterization (KickStart Log):** Device 2 displayed identical conduction behavior, achieving $26\ \mu\text{A}$ at 10 V. The slight variance between devices reflects local variations in contact area and microscale probe landing pressure.
- **Conduction Regime & Contact Physics:** The post-RTA I–V curve exhibits a quasi-ohmic, weakly non-linear turn-on below 2 V that transitions into linear channel conduction at higher bias. This behavior is characteristic of planar Al/Cr/Au contacts where quantum mechanical field emission through the thin $\text{AlN}/\text{V}_\text{N}$ interfacial barrier governs carrier injection into the 2DEG.



(a) Post-RTA I–V curve: Device 1 ($48\ \mu\text{A}$ @ 10 V)



(b) Post-RTA I–V curve: Device 2 ($26\ \mu\text{A}$ @ 10 V)

Figure 7: (a) Post-RTA Keithley KickStart I–V characteristic for Device 1 demonstrating active conduction up to $48\ \mu\text{A}$ at 10 V.

(b) Post-RTA characteristic for Device 2 reaching $26\ \mu\text{A}$ at 10 V.

11. Quantitative Comparative Analysis: Pre-RTA vs. Post-RTA

Table 4: Comprehensive Quantitative Comparison of Pre-RTA and Post-RTA Device Performance

Parameter / Metric	Pre-RTA (As-Deposited State)	Post-RTA (850°C , 45 s Annealed State)
Contact Metallurgical Phase	Unreacted Al / Cr / Au discrete thin films	AlN formation + dense V_N donors + Au-Al intermetallics
Conduction State	Open circuit / insulating blocking barrier	Active quasi-ohmic / ohmic carrier conduction
Drain Current @ 5 V	$< 5\ \text{nA}$ (Instrument noise / leakage floor)	$\sim 18\ \mu\text{A}$ (Device 1) / $\sim 10\ \mu\text{A}$ (Device 2)
Drain Current @ 10 V	Non-conductive ($\sim 0\ \mu\text{A}$)	$48\ \mu\text{A}$ (Device 1) / $26\ \mu\text{A}$ (Device 2)
Dominant Transport Mechanism	Thermionic emission suppressed by Φ_B	Field emission (quantum tunneling) via thin V_N layer into 2DEG